



PRECURSOR ANALYSIS REPORT: MUMBAI 2020 POWER OUTAGE – RELIABILITY FAILURE EXPOSES MALWARE INTRUSION

Cybersecurity for the Operational Technology
Environment (CyOTE)

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PRECURSOR ANALYSIS REPORT: MUMBAI 2020 POWER OUTAGE – RELIABILITY FAILURE EXPOSES MALWARE INTRUSION

1. EXECUTIVE SUMMARY

The Mumbai 2020 Power Outage – Reliability Failure Exposes Malware Intrusion Precursor Analysis Report leverages publicly available information about the 12 October 2020 power outage in Mumbai, India. While the outage itself was due to a series of reliability failures, the cyber investigation it prompted led to the discovery of a malware intrusion that held the grid's Information Technology (IT) and Operational Technology (OT) systems at risk, and about which the grid operators were completely unaware. This study catalogs anomalous observables for each technique employed in the intrusion, framed in the context of the reliability failures. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

The outage began with the failure of the four high-voltage transmission lines feeding power from the national transmission network into the Mumbai grid. One of the lines was already out of service, undergoing repairs after a conductor snapped on 10 October. A damaged insulator tripped a second line over-voltage at around 4:30 AM on 12 October, not long before the city's 2,500 electric commuter trains began their morning rush hour service. While grid operators called for additional hydro generation from Mumbai's embedded capacity early in the 9:00 AM hour, over half of that generation tripped roughly 45 minutes later due to a fault at a hydro station. This loss overloaded the third high voltage transmission line, which tripped at about 10:00 AM. Shortly thereafter, operators manually tripped the final incoming transmission line after they observed the line sparking due to a voltage overload at one of the substations. This led to the triggering of Mumbai's islanding scheme, which isolates the local grid and relies on embedded generation to keep the city's grid functioning in case of fluctuations in or failure of the national grid. Unfortunately, Mumbai's embedded generation capacity fell well short of the existing grid demand, tripping the city's power stations and sending the city into a blackout.^{1,2,3}

While investigations confirmed the outage was indeed due to reliability issues, Maharashtra state power officials remained concerned about the possibility of a cyber attack in light of recent regional tensions with China and commissioned the state police cyber authorities to conduct an investigation. The resulting report, released to the state government in March 2021, revealed that the IT and OT systems of the state's grid control systems had been compromised, likely by Chinese state-sponsored adversaries.^{4,5} The report claimed the adversaries were able to "easily" breach multiple firewalls and implant at least 14 trojans in both the IT and OT environments of the control networks governing the state's power distribution system. In addition, central government officials revealed that cyber attacks had been carried out against other segments of India's grid infrastructure at the time but claimed they were not linked to Mumbai's grid failure.

Researchers and analysts identified 14 unique techniques related to the malware most likely utilized for the intrusion with a total of 183 observables using MITRE ATT&CK® for Industrial Control Systems. The CyOTE program assesses observables accompanying techniques to identify opportunities to detect malicious activity. If observables accompanying the techniques are perceived and investigated during an intrusion or prior to the triggering event of a cyber attack, earlier comprehension of malicious activity can take place.

While the cyber intrusion against Mumbai's power grid never resolved into an actual attack, CyOTE analysts used the publicly available data on the intrusion to create a cyber attack scenario depicted by the notional attack sequence found in [Section 3](#). In this scenario, 11 of the identified techniques were precursors to the notional triggering event. Analysis identified 133 observables associated with these precursor techniques, 86 of which were assessed to have an increased likelihood of being perceived in the 160 days preceding the triggering event. The response and comprehension time could have been reduced if the observables had been identified earlier.

The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack. Organizations can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in Figure 1. CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.

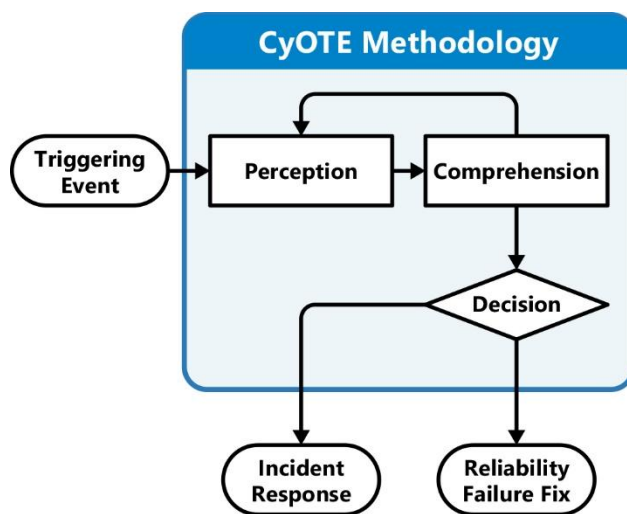


Figure 1. CyOTE Methodology

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the precursor analysis report's cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack.

2.2. BACKGROUND: RELIABILITY FAILURE

The metropolis of Mumbai, India, suffered a blackout from reliability failures at about 10:05 AM on 12 October 2020 (D-0) that paralyzed the city's 2,500 electric trains, stranding thousands of commuters,⁶ and left some hospitals, the Bombay Stock Exchange (BSE), and most of the city's 21 million inhabitants without power. While utilities began to restore power within an hour, most of the city was without electricity for over three hours, and power was not fully restored until 1:00 AM (H+15) the following morning.^{7,8,9}

As shown in Figure 2,¹⁰ 12 October began with the Mumbai grid already under some degree of stress: of the four 400 kV lines that provided national grid power to the city through the receiving station at Kalwa, which is Maharashtra's State Load Dispatch Center (SLDC), one had been out of service since the afternoon of 10 October due to a snapped conductor.

A second line tripped at around 4:30 AM on 12 October (H-5) on over-voltage due to a damaged insulator. That line was taken out of service just before 7:00 AM (H-3) to replace it and was projected to be back in service by 9:30 AM (M-35). However, that line was not restored until 10:41 AM (M+36), over 30 minutes after the power failure began.^{11,12}

The two remaining 400 kV lines, in addition to the city's internal generation capacity, were left carrying the entire load to support Mumbai's 21 million people just as rush hour began for the nearly 2,500 electric trains that carry over 7.5 million commuters across the city each day.^{13,14}

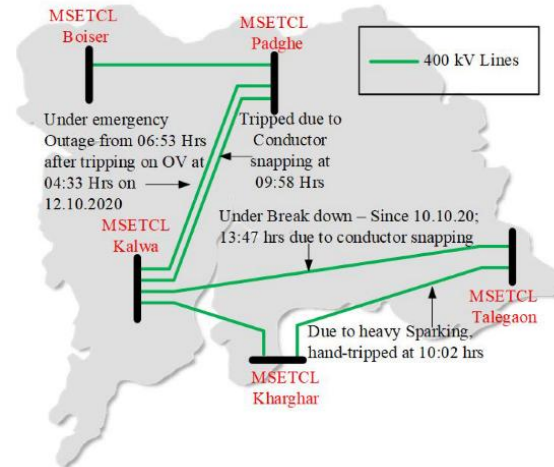


Figure 2. Status of 400 kV lines Serving Mumbai (Source: IEEE Xplore)

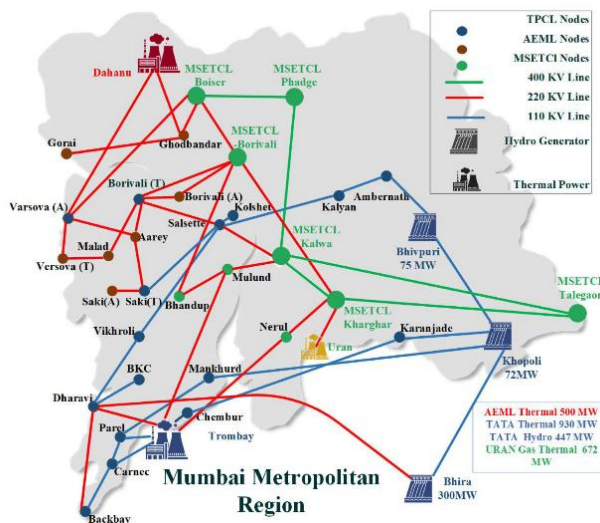


Figure 3. Mumbai's Main Power Grid (Source: IEEE Xplore)

Grid operators responded to the situation by calling for additional regional hydro generation at approximately 9:00 AM (H-1) to reduce the load on the two remaining 400 kV lines. Unfortunately, over half the additional hydro power tripped from a fault at the hydro station at 9:45 AM (M-20).^{15,16}

At approximately 10:00 AM (M-5) a third 400 kV line, carrying a load of 633 MW, tripped, which led to overloading on the fourth and final 400 kV line. The fourth line was manually tripped by operators after they saw sparks from the line at the intermediate Kharghar substation.^{17,18}

The tripping of the fourth incoming 400 kV line led to backfeeding from the Mumbai grid at approximately 10:05 AM (D-0) into the national grid which, combined with a drop in the grid frequency, triggered Mumbai's islanding mode scheme: the interconnects with the national grid

frequency, triggered Mumbai's islanding mode scheme: the interconnects with the national grid

tripped, isolating the city's grid with power provided by embedded generation capacity (Figure 3).^{19,20,a}

However, the city's embedded generation capacity could not support the demand, leading to the associated power generation stations tripping and sending the city into a blackout, with a total load loss of 3,508 MW and total generation loss of 1375 MW.²¹

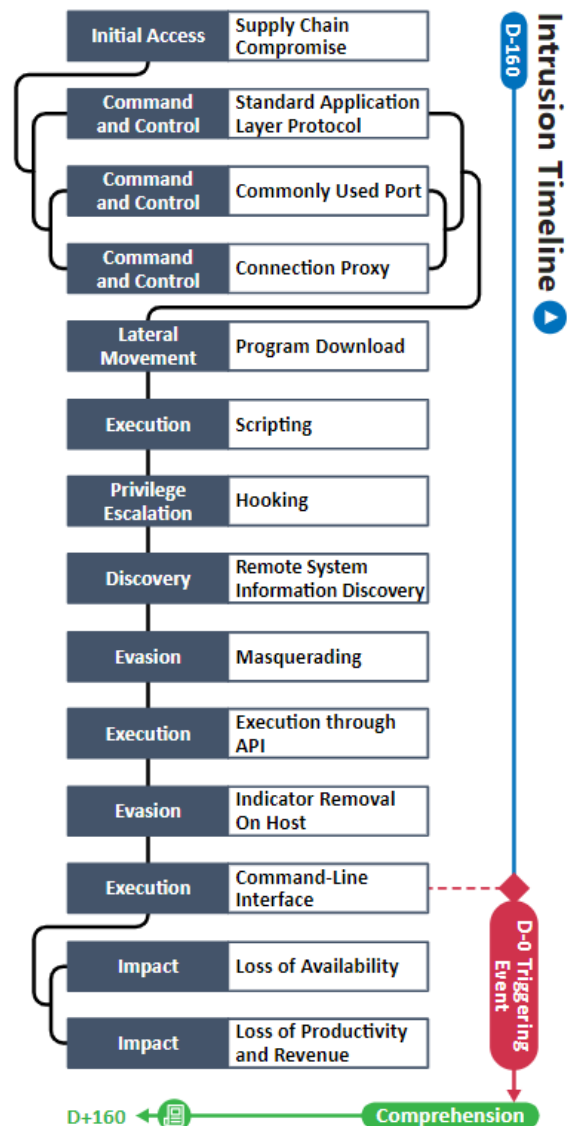
2.3. BACKGROUND: MALWARE INTRUSION

Maharashtra State Power Minister Nitin Raut was not convinced the 12 October 2020 (D-0) outage was due to a reliability failure. He believed the 400 kV supply through Kharghar – the line that operators manually tripped at about 10:00 AM (M-5), leading to the outage – should not have had to be shut down and that the city's grid should not have had to go into islanding mode. Raut suspected sabotage, particularly from a cyber attack, and his concerns prompted an investigation by the state police's Maharashtra Cyber Cell that began on 14 October (D+2).²²

Raut's concerns almost certainly stemmed from the context of India's relations at the time with China, which had hit a tragic 45 year low with a series of clashes between Indian and Chinese troops in the contested Ladakh region in northern India that began on 5 May of that year and continued through early September.²³

Coincidentally, cybersecurity firm Recorded Future observed a major spike in command and control (C2) activity associated with ShadowPad malware employed by China-affiliated adversaries against Indian infrastructure targets, particularly in the power sector. The spike began in early May, coinciding with the initial border clashes (D-160).²⁴ One of the ShadowPad targets was the Western Regional Load Dispatch Center (WRLDC) in Mumbai, under which falls the SLDC at Kalwa.

A timeline of adversarial techniques is shown in **Error! Reference source not found.**. The timeline includes the estimated number of days prior to and after the notional triggering event on 12 October that is depicted in the cyber attack scenario in Section 3. The timeline after the triggering event includes the assessed victim comprehension timeline.



^a Islanding, or island mode, is when a local grid such as Mumbai's cuts itself off from the national grid due to power fluctuations or other factors and continues in isolation using embedded generation capacity.

Recorded Future released its report one day before the preliminary report by the Maharashtra

Cyber Cell was turned over to state authorities on 1 March 2021 (D+160), which is assessed as the victim's full comprehension of the cyber intrusion.²⁵

Figure 4. Intrusion Timeline

While the Cyber Cell's report was not made public, Raut informed the press after reading it that the Mumbai outage was the result of "an organized cyber attack" against Maharashtra State Electricity Transmission Co., Ltd. (MSETCL), which happens to be the largest government-owned electric power transmission utility in India.²⁶ Raut then recounted several critical details from the report:^{27,28,29,30,31,32,33}

- Repeated attempts, at least some of which were successful, were made from blacklisted and suspicious IP addresses to log in to the Kalwa SLDC and Maharashtra State Electricity Board (MSEB)/MSETCL servers. These IP addresses, located in China, the UK, and other unspecified locations, had been certified as suspicious and malicious by unspecified major credit rating agencies.
- Fourteen trojans, type unspecified, were introduced into the MSEB/MSETCL computer systems. These trojans "easily" bypassed the firewalls of the Information Technology (IT) and Operational Technology (OT) servers that oversaw the electrical power distribution system. Some of these trojans had been used in previous cyber attacks in other parts of the world.^b
- The firewall of the Kalwa SLDC server was easily accessed by malicious code and software programs that could compromise the cybersecurity ecosystem, as well as disrupt its design functionality.
- Eight trojans were infiltrated through a compromised firewall into the Kalwa SLDC's Supervisory Control and Data Acquisition (SCADA) unit that balances supply and demand load dispatch.
- Roughly 8 GB of anomalous data was infiltrated into the Kalwa SLDC server through the previously mentioned suspicious IPs.^c
- Prior to the grid failure, the IT system at Kalwa SLDC sounded an alarm for a possible cyber attack three times in the span of a minute, but the alarms went unnoticed by the operators.
- The Cyber Cell made several key recommendations, including separating the IT and OT infrastructure, [improving] password management, updating web application security systems, updating the IT and OT design, and empowering the SLDC system.

After the revelations of the Cyber Cell report, Union Power Minister R.K. Singh from the central government maintained the outage was due to human error and not a cyber attack. However, he revealed that one of the teams that investigated the outage confirmed that cyber attacks had indeed taken place, but Singh insisted they were not linked to the Mumbai grid failure. Instead, these cyber attacks had targeted the Northern and Southern RLDCs, but the "malware could not reach the operating system."³⁴ Further details were not available, although the attacks Singh mentioned likely were a reflection of the intrusion activity observed by Recorded Future.

Analysis identified 14 unique techniques in a sequence and timeframe likely used by adversaries during the malware intrusion that are used in the cyber attack scenario in Section 3 (Table 1). These attack techniques are defined according to MITRE's ATT&CK[®] for ICS framework.

^b Raut provided no elaboration on the specifics of the malware, but it is highly likely that at least one of the trojans referred to was ShadowPad; the Cyber Cell may also have identified multiple instances of the same trojan in different parts of the compromised systems.

^c Some sources for Raut's statements indicated the data had been removed from the server, but the more consistent theme was that the data was introduced to it.

Table 1. Techniques Used in the Mumbai 2020 Malware Intrusion and Cyber Attack Scenario

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Data Historian Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Drive-by Compromise	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Engineering Workstation Compromise	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Default Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
Exploit Public-Facing Application	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Exploitation of Remote Services	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
External Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Internet Accessible Device	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Remote Services	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Replication Through Removable Media	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Rogue Master							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Spearphishing Attachment									Rootkit		Manipulation of View
Supply Chain Compromise									Service Stop		Theft of Operational Information
Wireless Compromise									System Firmware		

Table 2. Precursor Analysis Report Quantitative Summary

Precursor Analysis Report Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	14
Technique Observables	183
Precursor Techniques	11
Precursor Technique Observables	133
Highly Perceivable Precursor Technique Observable	86

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis. In the case of this precursor analysis report, the focus is on the core functionality of the ShadowPad malware, since no public information is available on any specific plugins that might have been employed in the intrusion into Mumbai's power grid.

3.1. SUPPLY CHAIN COMPROMISE TECHNIQUE (T0862) FOR INITIAL ACCESS

CyOTE analysts assess that ShadowPad malware was used in the intrusion that compromised Mumbai's grid control networks and systems.³⁵ The initial access was almost certainly established by early May 2020, if not before, as the adversary was actively using ShadowPad in various campaigns as early as 2017.³⁶ In the case of the Mumbai intrusion, the ShadowPad shellcode was embedded in legitimate .dynamic link libraries (.dll) within software packages produced by a software supplier. The victims may have witnessed an anomalous .dll file embedded in legitimate software, although the .dll is digitally signed and would not stand out as being malicious. ShadowPad is automatically initialized by C runtime code, which provides an adversary with remote access through a hidden backdoor.³⁷

IT Cybersecurity personnel may have been able to observe the download and installation of a compromised library.

A total of 15 observables were identified with the use of the [Supply Chain Compromise technique \(T0862\)](#). This technique is important for investigation because it presents a unique opportunity for an adversary to gain access into a victim's environment. This technique appears first in the timeline and responding to it would effectively halt adversary access to victim environments. Terminating the chain of techniques at this point would prevent the malware from infecting victims and limit operational damage in both IT and OT environments. This technique modifies the host operating system files via the injection of ShadowPad shell code into one or more software update packages installed by the victim, placing the code base into a modified or compromised state. If system backups are created after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the 15 observables associated with this technique, none are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 31 artifacts could be generated by the Supply Chain Compromise technique
Technique Observers^d	IT Cybersecurity
Resources	Technique Detection References

^d Observer titles are adapted from the Job Role Groupings listed in [the SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

3.2. STANDARD APPLICATION LAYER PROTOCOL TECHNIQUE (T0869) FOR COMMAND & CONTROL

Once the victim installed the ShadowPad-infected software, the adversary leveraged the Standard Application Layer Protocol technique to establish a connection between the C2 server and the victim machine(s). The Uniform Resource Locator (URL) for the C2 server had a default value of `dns://www.notped[.]com` but could vary if the adversary chose to specify a different URL.³⁸

The malware most likely used Hypertext Transfer Protocol (HTTP) proxied over port 8080 and Domain Name System (DNS) over port 53 to establish the connection. Depending on the URL scheme, File Transfer Protocol (FTP) or Hypertext Transfer Protocol Secure (HTTPS) over port 443 with GET requests could also be employed to reach an intermediary server. ShadowPad sends DNS requests every eight hours over Port 53 to reestablish the connection with the C2 server. A XOR-based encryption algorithm encrypts the request buffer, with the contents of the request buffer varying depending on the adversary's specifications. This request buffer is then converted to a readable string of Latin characters by adding each half of the byte to the 'a' and 'j' characters, respectively.³⁹

IT Staff and IT Cybersecurity personnel may have been able to observe various network traffic related to the employment of the Standard Application Layer Protocol technique, such as DNS, FTP, HTTP, and/or HTTPS traffic.

A total of 30 observables were identified with the use of the [Standard Application Layer Protocol technique \(T0869\)](#). This technique is important for investigation because it enables adversarial C2 capabilities and limits the ability of defenders to detect adversarial activity. This technique appears early in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at this point would prevent any further intrusion by the ShadowPad malware.

Of the 30 observables associated with this technique, 22 are assessed to be highly perceivable (Outbound Connections to Anomalous External IPs; DNS Connections to Anomalous Domain; DNS Connections to Anomalous Domain: `dns://www.notped[.]com`; Anomalous DNS Traffic; Anomalous DNS Traffic Over UDP Port 53 to External Hosts; Anomalous DNS Traffic with Requests Over UDP Port 53 to External Hosts; Anomalous DNS Traffic with Requests Over UDP Port 53 to External Hosts Every 8 hours; Anomalous Outbound HTTP Traffic Over TCP Port 8080 to External Hosts; Anomalous Outbound FTP Traffic; Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts; Anomalous Inbound File Transfers from External Host; Anomalous Inbound DNS Traffic; Anomalous Inbound DNS Traffic Containing Encrypted Strings; Anomalous DNS Requests; Anomalous DNS Requests to Dynamically Changing Domains; Anomalous DNS Requests to Dynamically Changing Domains: `"*.....*.%DGA-domain%.com"`; Anomalous DNS Requests to Dynamically Changing Domains: `"dns://www.notped[.]com"`; Anomalous DNS Requests Sent to Public DNS Resolvers; Anomalous DNS Requests Sent to IP Address 4.2.2.1; Anomalous DNS Requests Sent to IP Address 4.2.2.2; Anomalous DNS Requests Sent to IP Address 8.8.8.8; Anomalous DNS Requests Sent to IP Address 8.8.4.4).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Standard Application Layer Protocol technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.3. COMMONLY USED PORT TECHNIQUE (T0885) FOR COMMAND & CONTROL

As a part of the aforementioned Standard Application Layer Protocol technique, the adversary used standard ports for most of these protocols. Depending on the protocol specified for C2, communications would occur over commonly used ports such as 443 for HTTPS and 53 for DNS. HTTP proxying occurred over port 8080, which while not the standard HTTP port, is not uncommon.⁴⁰

IT Staff and IT Cybersecurity personnel may have been able to observe intrusion-related network traffic – such as DNS, FTP, HTTP, and/or HTTPS – over common ports.

A total of eight observables were identified with the use of [the Commonly Used Port technique \(T0885\)](#). This technique is important for investigation because it enables command and control of the host environment by the adversary and limits the ability of defenders to detect adversarial activity. This technique appears early in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at this point would prevent any further intrusive action from the attacker using the ShadowPad malware.

Of the eight observables associated with this technique, five are assessed to be highly perceivable (Anomalous DNS Traffic; Anomalous DNS Traffic Over UDP Port 53 to External Hosts; Anomalous Outbound HTTP Traffic to External Proxy Over TCP Port 8080; Anomalous Outbound FTP Traffic; Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of five artifacts could be generated by the Commonly Used Port technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.4. CONNECTION PROXY TECHNIQUE (T0884) FOR COMMAND AND CONTROL

At this point, the malware generates a domain name for the C2 server, in which a hash table generates a domain name based on the date. The victim host contacts the C2 server via the generated domain over FTP (Port 20 or 21), HTTP (Port 8080), or HTTPS (Port 443) to retrieve information regarding the address of the primary C2 server. Using this information, the actual name of the primary C2 server is created using a domain generation algorithm (DGA) based on a mask specified in the configuration data, such as prefix%DGA-generated-part%suffix, and the location of the suffix is marked with an '@' character.⁴¹

IT Staff and IT Cybersecurity personnel may have been able to observe DNS, FTP, HTTP, and/or HTTPS traffic being employed to transmit information related to the address of the primary C2 server as well as the proxying of HTTP over a non-standard port.

A total of 14 observables were identified with the use of the [Connection Proxy technique \(T0884\)](#). This technique is important for investigation because it enables command and control of the host environment by the adversary and limits the ability of defenders to detect adversarial activity. This technique appears early in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at this point would prevent any further intrusive action from the attacker using the ShadowPad malware.

Of the 14 observables associated with this technique, 10 are assessed to be highly perceivable (Anomalous Use of DNS Privacy Services; Anomalous Use of DNS Privacy Services; WhoisProtection WHOIS Privacy Protection Service; Anomalous HTTPS Traffic Over Port 8080; Anomalous HTTPS Traffic Over Port 8080 to External Hosts; Anomalous HTTPS Traffic Over Port 8080 to External Hosts to Spoofed Domains (Spoofed Domains Look Similar to victim Entities); Anomalous HTTPS Traffic Over Port 443 to External Hosts to Spoofed Domains (Spoofed Domains Look Similar to Victim Entities); Anomalous Outbound HTTP Traffic to External Proxy; Anomalous Outbound HTTP Traffic to External Proxy Over TCP Port 8080; Anomalous Outbound FTP Traffic; Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of six artifacts could be generated by the Connection Proxy technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.5. PROGRAM DOWNLOAD TECHNIQUE (T0843) FOR LATERAL MOVEMENT

After establishing a connection with a C2 server, ShadowPad acts as a modular backdoor platform. If the backdoor is activated, the adversary can upload files, create processes, and store information in a virtual file system (VFS) contained within the victim machine's registry. ShadowPad's download functionality also allows the adversary to download and execute specialized ShadowPad plugins and any additional code they might wish to employ against the target network.⁴² ShadowPad employs outbound HTTP/HTTPS/FTP traffic to initiate C2 server communication and coinciding inbound HTTP/HTTPS/FTP traffic to download plugins. If logging is enabled, these events may generate system event logs and firewall alerts.

IT Staff and IT Cybersecurity personnel may have been able to observe anomalous HTTP requests to and file transfers from external hosts.

A total of 20 observables were identified with the use of the [Program Download technique \(T0854\)](#). This technique is important for investigation because it enables unauthorized access to victim assets and modifies the host operating environment. This technique appears relatively early in the timeline and responding to it will disrupt the adversary's intrusion progression. Terminating the chain of techniques at this point would prevent the use of additional malware plugins and disrupt the progression of the adversary's attack chain, thereby limiting operational damage. This technique modifies the host operating system files via the creation of anomalous services and modification of host operating system files, placing the host into a modified or compromised state. If system backups are created after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the 20 observables associated with this technique, 16 are assessed to be highly perceivable (Anomalous DNS Traffic; Anomalous DNS Traffic Over UDP Port 53 to External Hosts; Anomalous Outbound HTTP Traffic Over TCP Port 8080 to External Hosts; Anomalous Outbound FTP Traffic; Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts; Anomalous Inbound HTTP Traffic; Anomalous Inbound HTTP Traffic Over TCP Port 8080 from External Hosts; Anomalous Inbound HTTPS Traffic; Anomalous Inbound HTTPS Traffic Over TCP Port 443 from External Hosts; Anomalous Inbound FTP Traffic; Anomalous Inbound FTP Traffic Over TCP port 20 and 21 from External Hosts).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 19 artifacts could be generated by the Program Download technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.6. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

Once ShadowPad downloads any designated plugins or other code, scripting is employed to initialize them. Initialization of the plugin involves extensive use of scripting, as does implementing specific functionality of certain plugins. A plugin is initiated by calling sequentially its entry point, Dllmain, with Reason parameters 101, 102, and 104; the returned data is then copied in a structure describing the plugin. Once the plugin returns no error during initialization, it is added to ShadowPad's plugin list.⁴³

IT Staff and IT Cybersecurity personnel may have been able to observe privilege escalation or modification, or related service event entries in Windows event logs.

A total of four observables were identified with the use of the [Scripting technique \(T0853\)](#). This technique is important for investigation because it facilitates privilege escalation and enhances adversarial access into a victim's environment. This technique appears early in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at this point would disrupt the adversary's ability to effectively utilize additional ShadowPad plugins, thereby limiting operational damage. This technique modifies the host operating system files via the creation of anomalous processes and modification of host operating system files, resulting in the host being placed into a modified or compromised state. If system backups are created after this technique is executed, data recovery and disaster recovery efforts will be impaired.

All four observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Scripting technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.7. HOOKING TECHNIQUE (T0874) FOR PRIVILEGE ESCALATION

ShadowPad hooks into different processes depending on its configuration and further adjusts process privileges in order to escalate itself before invoking the Config plugin's function, LoadConfig.⁴⁴

IT Staff and IT Cybersecurity personnel may have been able to observe the creation of anomalous processes associated with ShadowPad or its plugins, as well as anomalous privilege changes.

A total of 14 observables were identified with the use of the [Hooking technique \(T0874\)](#). This technique is important for investigation because it facilitates privilege escalation and enhances adversarial access into a victim's environment. This technique appears relatively early in the timeline and responding to it will effectively halt all future events. Terminating the chain of techniques at this point would reduce the adversary's ability to operate discretely while also preventing them from utilizing their malicious toolset, effectively limiting operational damage. This technique modifies the host operating system files via the creation of anomalous processes and writing of anomalous files, resulting in the host being placed into a modified or compromised state. If system backups are created after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the 14 observables associated with this technique, eight are assessed to be highly perceivable (Anomalous Process Privilege Change (Windows Event ID 4673, 4674); Anomalous Program Files; Anomalous .DLL files; Anomalous .DLL files: TSVIPsrv.DLL; Files Executed from Privileged Directory; Files Executed from Windows System32 Directory; Anomalous Files Executed from Windows System32 Directory; Anomalous .DLLs Executed from Windows System32 Directory).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 11 artifacts could be generated by the Hooking technique
Technique Observers	IT Staff, IT Cybersecurity
Resources	Technique Detection References

3.8. REMOTE SYSTEM INFORMATION DISCOVERY TECHNIQUE (T0888) FOR DISCOVERY

The primary C2 server reaches out to the infected machine to request information about it, such as current date and time, memory status, central processing unit (CPU) frequency, amount of free disk space, video mode, system locale, process ID (PID) of the malicious process, operating system (OS) version, domain name, and username. The information gathered is then copied to configuration storage and updates the corresponding registry key.⁴⁵

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous file transfers and requests (FTP and HTTP) between the infected host and the C2 server.

A total of eight observables were identified with the use of the [Remote System Information Discovery technique \(T0888\)](#). This technique is important for investigation because it enables data extraction as well as persistence. This technique appears in the middle of the timeline and responding to it may allow defenders to prevent data exfiltration, as well as further system modification and data loss. Terminating the chain of techniques at this point would limit operational damage as the adversary would most likely not have enough information on the target devices to effectively carry out an attack.

Of the eight observables associated with this technique, five are assessed to be highly perceivable (Anomalous Outbound HTTP Traffic Over TCP Port 8080 to External Hosts; Anomalous Inbound HTTP Traffic; Anomalous Inbound HTTP Traffic Over TCP Port 8080 from External Hosts; Anomalous Inbound HTTPS Traffic; Anomalous Inbound HTTPS Traffic Over TCP Port 443 from External Hosts).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of eight artifacts could be generated by the Remote System Information Discovery technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.9. MASQUERADING TECHNIQUE (T0849) FOR EVASION

Depending on its configuration, ShadowPad can inject itself into a wide variety of DLLs and processes and then masquerade as those processes. After initial setup, the legitimate executable is launched as a Windows service, which in turn initiates ShadowPad's execution chain. The chain begins with the execution of ShadowPad (C:\Users\Desktop\log.exe). Then a service is created (SERVICE C:\ProgramData\Microsoft\DEV\Scripts\reg.exe). The third step is service execution, where the service created in the previous step (i.e., ...reg.exe) is executed. The final step is payload injection in the compromised network, in this case C:\Windows\system32\svchost.exe.⁴⁶

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe the Windows service creating unusual processes, such as svchost.exe, and the spawning of multiple dllhost.exe and cmd.exe child processes.⁴⁷

A total of eight observables were identified with the use of the [Masquerading technique \(T0849\)](#). This technique is important for investigation as the adversary purposefully disguises files so staff may not suspect malicious applications or executables. This technique appears in the middle of the timeline and responding to it would enhance detection of anomalies by defenders. This technique modifies the host operating system files via the manipulation of system configuration settings and associated system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the eight observables associated with this technique, seven are assessed to be highly perceivable (Files Executed from Privileged Directory; Files Executed from Windows System32 Directory; Anomalous Files Executed from Windows System32 Directory; Anomalous .DLLs Executed from Windows System32 Directory; Anomalous Program Files; Anomalous .DLL files; Anomalous .DLL files: TSVIPSrv.DLL).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 15 artifacts could be generated by the Masquerading technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.10. EXECUTION THROUGH API TECHNIQUE (T0871) FOR EXECUTION

ShadowPad's Plugins module provides Application Programming Interface (API) functionality. At a minimum, ShadowPad would require this functionality to import other plugins, and at least some plugins would require API functionality for external activation. The Plugins module also provides an API for other modules but does not initiate any actions without external intervention by the adversary. The Plugins module initiates a registry monitoring thread and waits for any changes in its registry key. The module then loads any new plugins it finds into the registry VFS (HKLM\HKCU\SOFTWARE\Microsoft\%5-12 random characters%).⁴⁸

Every value found in the key is decrypted and checked to determine if it is a valid plugin, after which it is loaded and initialized using the API from ShadowPad's Root module. The Plugins module also provides an API for reading, writing, and deleting arbitrary register values, which also enables the writing of new plugin images to the registry VFS using commands from the C2 server.⁴⁹

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe the creation of anomalous processes and registry keys.

A total of eight observables were identified with the use of the [Execution Through API technique \(T0871\)](#). This technique is important for investigation because it enhances the ability of malware to compromise victim operational assets and enhances the ability of the adversary to maintain control over them. This technique appears in the middle of the timeline and responding to it will disrupt the adversary's ability to carry out their attack. Terminating the chain of techniques at this point would limit operational damage by negating the adversary's ability to affect change(s) against target devices. This technique modifies the host operating system files via the manipulation of system configuration settings and associated system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the eight observables associated with this technique, six are assessed to be highly perceivable (Network Traffic Captures Contain Anomalous Binaries; Network Traffic Captures Contain Anomalous Function Code Related to ShadowPad Plugins; Anomalous Object Deletion (Windows Event ID: 4660); Anomalous Registry Value Was Modified (Windows Event ID: 4657); Presence of Anomalous Registry Keys; Presence of Anomalous Registry Keys: HKLM\HKCU\Software\Microsoft\<key>).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 19 artifacts could be generated by the Execution Through API technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.11. INDICATOR REMOVAL ON HOST TECHNIQUE (T0872) FOR EVASION

Similar to masquerading, ShadowPad uses yet another technique in an attempt to evade discovery: any files ShadowPad creates are stored on the host device in an encrypted format and in a location unique to each victim. This includes the VFS and any other files ShadowPad creates.⁵⁰

IT Staff, IT Cybersecurity, OT Staff, and OT Cybersecurity personnel may have been able to observe the creation or existence of anomalous encrypted files, as well as creation or deletion of files.

A total of four observables were identified with the use of [the Indicator Removal on Host technique \(T0872\)](#). This technique is important for investigation because it limits detection and recovery of victim assets by defenders. This technique appears in the middle of the timeline and responding to it will potentially disrupt the ShadowPad malware's ability to run as intended, effectively halting all future events. Terminating the chain of techniques at this point would limit operational damage. This technique modifies the host operating system files via the manipulation of system configuration settings and associated system registry entries, placing the host into a modified or compromised state. If system backups occur after this technique is executed, data recovery and disaster recovery efforts will be impaired.

Of the four observables associated with this technique, three are assessed to be highly perceivable (Presence of Anomalous Encrypted Files; Anomalous File Deletion (Windows Event ID: 4660); Anomalous Object Deletion (Windows Event ID: 4660)).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the Indicator Removal on Host technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.12. COMMAND-LINE INTERFACE TECHNIQUE (T0807) FOR EXECUTION

The extent to which the Mumbai grid control networks were compromised, based on the information State Power Minister Raut revealed in early March 2021, strongly suggests the adversary employed the Command-Line Interface technique (T0807) for at least some aspects of the intrusion. The adversary can spawn command-line instances with ShadowPad's base functionality, although this is not necessarily something that happens in an automated fashion.⁵¹

CyOTE analysts assess that the adversary would likely employ this technique to execute the necessary commands for ShadowPad and any other malicious code to take down the victim systems and networks, resulting in the triggering event (D-0).

IT Staff, IT Cybersecurity, OT Staff, and OT Cybersecurity personnel may have been able to observe anomalous terminal windows or creation of anomalous processes.

A total of nine observables were identified with the use of the [Command-Line Interface technique \(T0807\)](#). This technique is important for investigation because it facilitates malicious code execution and enables subsequent deployment of follow-on payloads and components. This technique appears later in the timeline and responding to it will likely disrupt the adversary's ability to utilize additional ShadowPad plugins. Terminating the chain of techniques at this point would limit the adversary's ability to continue the attack and thereby limit operational damage.

Of the nine observables associated with this technique, four are assessed to be highly perceivable (Appearance of Anomalous Command-Line Terminal at Startup; Anomalous Process Creation (Windows Event ID: 4688); Presence of Anomalous Application; Presence of Application with Anomalous Hash).

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 25 artifacts could be generated by the Command-Line Interface technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity
Resources	Technique Detection References

3.13. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT

Considering the extent to which the adversaries had compromised Mumbai's grid control networks, particularly the SCADA unit for balancing supply and load dispatch at Kalwa SLDC, CyOTE analysts assess they likely had the ability to initiate a blackout of Mumbai similar to what took place on 12 October 2020. They also likely could have hampered recovery efforts of the grid operators by manipulating and, through encryption or with disk wiper malware, destroying critical data and systems across the IT and OT environments.

All organizational personnel – including Engineering, IT Staff, IT Cybersecurity, Management, OT Staff, OT Cybersecurity, and Support Staff – would almost certainly have been able to observe the grid failure and the resulting loss of availability. In addition, IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity would likely have been able to observe warnings from intrusion alarms (as reportedly happened immediately prior to the actual blackout), execution of anomalous processes, and interruption or termination of legitimate processes.

A total of 17 observables were identified with the use of the [Loss of Availability technique \(T0826\)](#). This technique is important for investigation because it involves a direct loss of availability of IT and OT systems for the victim. Additionally, this technique may present an impact for the end users or consumers of products and services. This technique presents noticeable effects, including non-responsive devices, widespread power outages; interruption of business and other services; and costs to repair damaged infrastructure. This technique appears at the end of the timeline and responding to it will include efforts to regain operational functionality and resume normal operation.

All 17 observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of eight artifacts could be generated by the Loss of Availability technique
Technique Observers	Engineering, IT Staff, IT Cybersecurity, Management, OT Staff, OT Cybersecurity, and Support Staff
Resources	Technique Detection References

3.14. LOSS OF PRODUCTIVITY AND REVENUE TECHNIQUE (T0828) FOR IMPACT

The impact of a grid outage in a major metropolitan area is difficult to quantify, given that the loss of productivity and revenue do not fall mainly to the grid operator, but to the customers the grid serves. Using the 12 October 2020 outage in Mumbai as a reference for losses that likely would be suffered due to a cyber attack inflicting an outage lasting a full day, the direct revenue losses to MSETCL for a one-day outage^e would have been roughly \$1.8 million dollars, with additional losses suffered by the power generation companies that went off-line when the grid failed.^{52,53}

However, the far greater impact would be the losses inflicted on the entire city of Mumbai, which has the highest gross domestic product (GDP) of any Indian city. As of May 2022, Mumbai's contributions to the national economy were roughly \$384 billion annually.⁵⁴ Mumbai is also responsible for 25 percent of the country's industrial output, 70 percent of maritime trade, and 70 percent of capital transactions.⁵⁵ As such, a one-day outage would result in an estimated aggregate loss of just over \$1 billion in revenues, with corresponding losses in industrial output, maritime trade, and trading on the BSE.

All grid organization personnel – including Engineering, IT Staff, IT Cybersecurity, Management, OT Staff, OT Cybersecurity, and Support Staff – would almost certainly be able to observe the grid failure, and Management and Support Staff would have insight into the financial losses posed by such an outage to the grid operators.

A total of 24 observables were identified with the use of the [Loss of Productivity and Revenue technique \(T0828\)](#). This technique is important for investigation because it involves a direct loss of revenue and productivity for the victim. Additionally, this technique may present an impact for the end users or consumers of products and services. This technique appears at the end of the timeline and responding to it will include efforts to regain operational functionality and resume normal operation.

All 24 observables are assessed to be highly perceivable.

Please see [Appendix A](#) for the list of observables.

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of five artifacts could be generated by the Loss of Productivity and Revenue technique
Technique Observers	Engineering, IT Staff, IT Cybersecurity, Management, OT Staff, OT Cybersecurity, and Support Staff
Resources	Technique Detection References

^e The 12 October 2020 outage left most of the city without power for over three hours, with utilities restoring full service very early on the morning of 13 October.

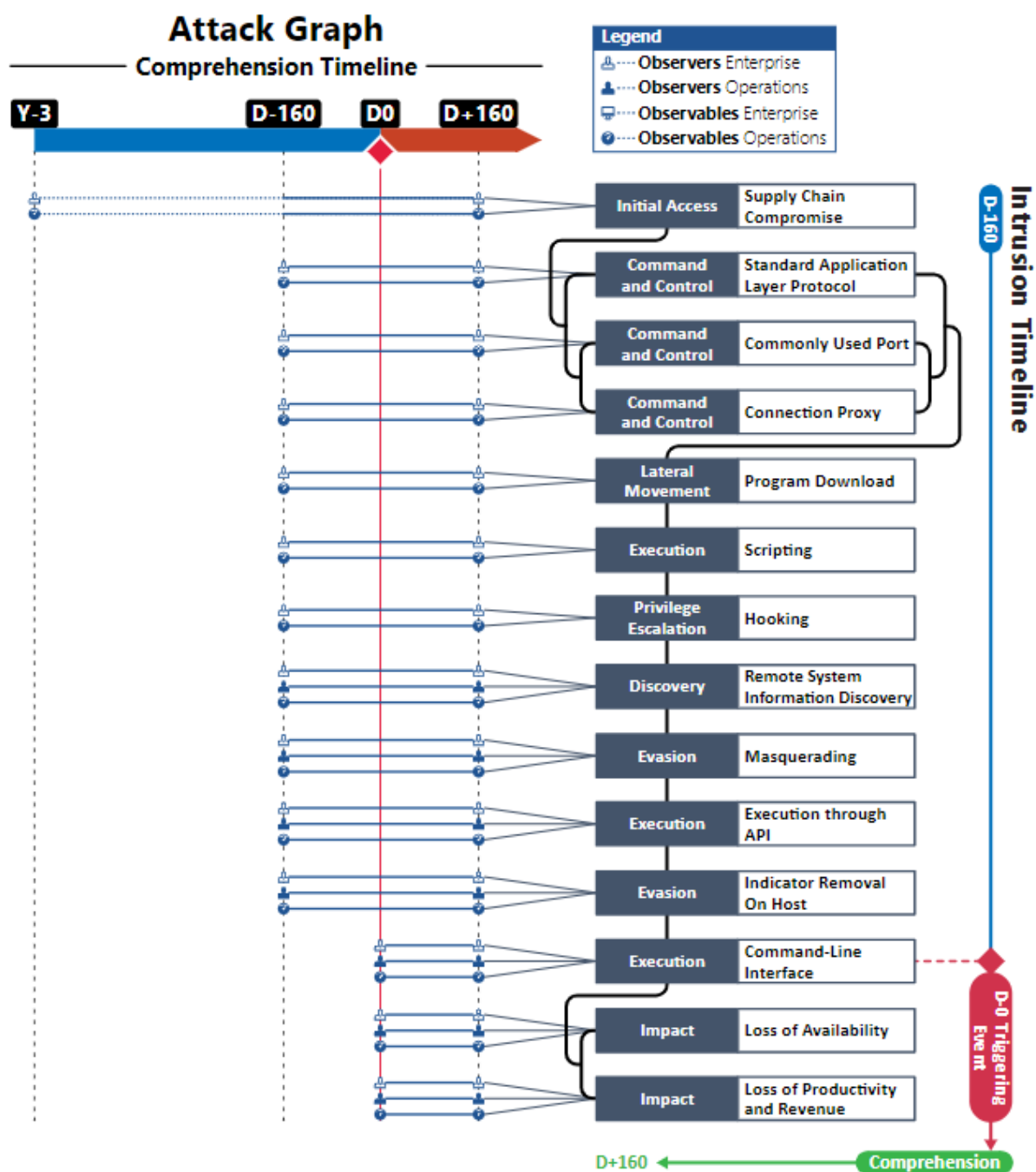


Figure 5. Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

NOTE: Highly perceivable observables are highlighted in italics

Observables Associated with Supply Chain Compromise Technique (T0862)	
Observable 1	Anomalous Software Embedded in Program Files
Observable 2	Anomalous Software Embedded in .DLLs
Observable 3	Anomalous Shellcode Embedded in Program Files
Observable 4	Anomalous Shellcode Embedded in .DLLs
Observable 5	Anomalous .DLL Embedded in Software
Observable 6	Anomalous Signatures on Anomalous Program Files
Observable 7	Anomalous Signatures on Anomalous .DLLs
Observable 8	Presence of Anomalous Runtime Functions
Observable 9	Presence of Anomalous C Runtime Functions
Observable 10	Presence of Anomalous C Runtime Function: <code>_Initterm(_PVFV * Pfbegin, _PVFV * Pfbegin);</code>
Observable 11	Anomalous Access Requests to Software Supplier Code Repositories
Observable 12	Anomalous Runtime Code
Observable 13	Anomalous C Runtime Code
Observable 14	Anomalous Runtime Code with Auto-Initialized Objects
Observable 15	Anomalous C Runtime Code with Auto-Initialized Objects

Observables Associated with Standard Application Layer Protocol Technique (T0869)	
Observable 1	Outbound Connections to External Ips
Observable 2	<i>Outbound Connections to Anomalous External Ips</i>
Observable 3	<i>DNS Connections to Anomalous Domain</i>
Observable 4	<i>DNS Connections to Anomalous Domain: dns://www.notped[.]com</i>
Observable 5	<i>Anomalous DNS Traffic</i>
Observable 6	<i>Anomalous DNS Traffic Over UDP Port 53 to External Hosts</i>
Observable 7	<i>Anomalous DNS Traffic with Requests Over UDP Port 53 to External Hosts</i>
Observable 8	<i>Anomalous DNS Traffic with Requests Over UDP Port 53 to External Hosts Every 8 Hours</i>
Observable 9	Anomalous Outbound HTTP Traffic
Observable 10	<i>Anomalous Outbound HTTP Traffic Over TCP Port 8080 to External Hosts</i>
Observable 11	<i>Anomalous Outbound FTP Traffic</i>
Observable 12	<i>Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts</i>
Observable 13	Anomalous Outbound HTTPS Traffic

Observables Associated with Standard Application Layer Protocol Technique (T0869)	
Observable 14	Anomalous Outbound HTTPS Traffic Over TCP Port 443 to External Hosts
Observable 15	Anomalous Outbound HTTPS GET Requests
Observable 16	Anomalous Outbound HTTPS GET Requests Over TCP Port 443 to External Hosts
Observable 17	<i>Anomalous Inbound File Transfers from External Host</i>
Observable 18	<i>Anomalous Inbound DNS Traffic</i>
Observable 19	<i>Anomalous Inbound DNS Traffic Containing Encrypted Strings</i>
Observable 20	<i>Anomalous DNS Requests</i>
Observable 21	<i>Anomalous DNS Requests to Dynamically Changing Domains</i>
Observable 22	<i>Anomalous DNS Requests to Dynamically Changing Domains: "*.....*.%DGA-Domain%.Com"</i>
Observable 23	<i>Anomalous DNS Requests to Dynamically Changing Domains: "dns://www.notped[.]com"</i>
Observable 24	<i>Anomalous DNS Requests Sent to Public DNS Resolvers</i>
Observable 25	<i>Anomalous DNS Requests Sent to IP Address 4.2.2.1</i>
Observable 26	<i>Anomalous DNS Requests Sent to IP Address 4.2.2.2</i>
Observable 27	<i>Anomalous DNS Requests Sent to IP Address 8.8.8.8</i>
Observable 28	<i>Anomalous DNS Requests Sent to IP Address 8.8.4.4</i>
Observable 29	DNS Resolves to Autonomous System Number (ASN)
Observable 30	DNS Resolves to Autonomous System Number (ASN) Number 3356 (Managed by Level-3 Parent LLC)

Observables Associated with Commonly Used Port Technique (T0885)	
Observable 1	<i>Anomalous DNS Traffic</i>
Observable 2	<i>Anomalous DNS Traffic Over UDP Port 53 to External Hosts</i>
Observable 3	Anomalous Outbound HTTP Traffic to External Proxy
Observable 4	<i>Anomalous Outbound HTTP Traffic to External Proxy Over TCP Port 8080</i>
Observable 5	<i>Anomalous Outbound FTP Traffic</i>
Observable 6	<i>Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts</i>
Observable 7	Anomalous Outbound HTTPS Traffic
Observable 8	Anomalous Outbound HTTPS Traffic Over TCP Port 443 to External Hosts

Observables Associated with Connection Proxy Technique (T0884)	
Observable 1	<i>Anomalous Use of DNS Privacy Services</i>

Observables Associated with Connection Proxy Technique (T0884)	
Observable 2	<i>Anomalous Use of DNS Privacy Services: Whoisprotection WHOIS Privacy Protection Service</i>
Observable 3	<i>Anomalous HTTPS Traffic Over Port 8080</i>
Observable 4	<i>Anomalous HTTPS Traffic Over Port 8080 to External Hosts</i>
Observable 5	<i>Anomalous HTTPS Traffic Over Port 8080 to External Hosts to Spoofed Domains (Spoofed Domains Look Similar to Victim Entities)</i>
Observable 6	<i>Anomalous HTTPS Traffic Over Port 443</i>
Observable 7	<i>Anomalous HTTPS Traffic Over Port 443 to External Hosts</i>
Observable 8	<i>Anomalous HTTPS Traffic Over Port 443 to External Hosts to Spoofed Domains (Spoofed Domains Look Similar to Victim Entities)</i>
Observable 9	<i>Anomalous Outbound HTTP Traffic to External Proxy</i>
Observable 10	<i>Anomalous Outbound HTTP Traffic to External Proxy Over TCP Port 8080</i>
Observable 11	<i>Anomalous Outbound FTP Traffic</i>
Observable 12	<i>Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts</i>
Observable 13	<i>Anomalous Outbound HTTPS Traffic</i>
Observable 14	<i>Anomalous Outbound HTTPS Traffic Over TCP Port 443 to External Hosts</i>

Observables Associated with Program Download Technique (T0843)	
Observable 1	<i>Anomalous DNS Traffic</i>
Observable 2	<i>Anomalous DNS Traffic Over UDP Port 53 to External Hosts</i>
Observable 3	<i>Anomalous Outbound HTTP Traffic</i>
Observable 4	<i>Anomalous Outbound HTTP Traffic Over TCP Port 8080 to External Hosts</i>
Observable 5	<i>Anomalous Outbound HTTPS Traffic</i>
Observable 6	<i>Anomalous Outbound HTTPS Traffic Over TCP Port 443 to External Hosts</i>
Observable 7	<i>Anomalous Outbound FTP Traffic</i>
Observable 8	<i>Anomalous Outbound FTP Traffic Over TCP Port 20 and 21 to External Hosts</i>
Observable 9	<i>Anomalous Inbound HTTP Traffic</i>
Observable 10	<i>Anomalous Inbound HTTP Traffic Over TCP Port 8080 from External Hosts</i>
Observable 11	<i>Anomalous Inbound HTTPS Traffic</i>
Observable 12	<i>Anomalous Inbound HTTPS Traffic Over TCP Port 443 from External Hosts</i>
Observable 13	<i>Anomalous Inbound FTP Traffic</i>
Observable 14	<i>Anomalous Inbound FTP Traffic Over TCP Port 20 and 21 from External Hosts</i>
Observable 15	<i>Anomalous Network Traffic to Anomalous External Hosts</i>
Observable 16	<i>Anomalous Network Traffic to Anomalous External Host: dns://noteped[.]com</i>
Observable 17	<i>Anomalous Network Traffic from Anomalous External Hosts</i>

Observables Associated with Program Download Technique (T0843)

Observable 18	Anomalous Network Traffic from Anomalous External Host: dns://noteped[.]com
Observable 19	Anomalous Plug-Ins Downloaded to Host
Observable 20	Anomalous ShadowPad Plug-Ins Downloaded to Host

Observables Associated with Scripting Technique (T0853)

Observable 1	<i>Anomalous Privilege Escalation (Windows Event ID 4673, 4674)</i>
Observable 2	<i>A Privileged Service Was Called (Windows Event ID: 4673)</i>
Observable 3	<i>An Operation Was Attempted on A Privileged Object (Windows Event ID: 4674)</i>
Observable 4	<i>Anomalous Modification of Registry Value: Environment\Userinitmplogonscript (Windows Event ID: 4657)</i>

Observables Associated with Hooking Technique (T0874)

Observable 1	Anomalous Process Creation (Windows Event ID: 4688)
Observable 2	<i>Anomalous Process Privilege Change (Windows Event ID 4673, 4674)</i>
Observable 3	<i>Anomalous Program Files</i>
Observable 4	<i>Anomalous .DLL Files</i>
Observable 5	<i>Anomalous .DLL Files: Tsviprv.DLL</i>
Observable 6	Anomalously Spawned Child Process
Observable 7	Anomalously Spawned Child Process from Common Executable
Observable 8	Anomalously Spawned Child Process from Common Executable: Svchost.Exe
Observable 9	Anomalously Spawned Child Process from Common Executable: Dllhost.Exe
Observable 10	Anomalously Spawned Child Process from Common Executable: Cmd.Exe
Observable 11	<i>Files Executed from Privileged Directory</i>
Observable 12	<i>Files Executed from Windows System32 Directory</i>
Observable 13	<i>Anomalous Files Executed from Windows System32 Directory</i>
Observable 14	<i>Anomalous .DLLs Executed from Windows System32 Directory</i>

Observables Associated with Remote System Information Discovery Technique (T0888)

Observable 1	Anomalous Outbound HTTP Traffic
Observable 2	<i>Anomalous Outbound HTTP Traffic Over TCP Port 8080 to External Hosts</i>
Observable 3	<i>Anomalous Inbound HTTP Traffic</i>
Observable 4	<i>Anomalous Inbound HTTP Traffic Over TCP Port 8080 from External Hosts</i>

Observables Associated with Remote System Information Discovery Technique (T0888)	
Observable 5	Anomalous Outbound HTTPS Traffic
Observable 6	Anomalous Outbound HTTPS Traffic Over TCP Port 443 to External Hosts
Observable 7	<i>Anomalous Inbound HTTPS Traffic</i>
Observable 8	<i>Anomalous Inbound HTTPS Traffic Over TCP Port 443 from External Hosts</i>

Observables Associated with Masquerading Technique (T0849)	
Observable 1	Anomalous Process Creation (Windows Event ID: 4688)
Observable 2	<i>Files Executed from Privileged Directory</i>
Observable 3	<i>Files Executed from Windows System32 Directory</i>
Observable 4	<i>Anomalous Files Executed from Windows System32 Directory</i>
Observable 5	<i>Anomalous .DLLs Executed from Windows System32 Directory</i>
Observable 6	<i>Anomalous Program Files</i>
Observable 7	<i>Anomalous .DLL Files</i>
Observable 8	<i>Anomalous .DLL files: TSVIPsrv.DLL</i>

Observables Associated with Execution Through API Technique (T0871)	
Observable 1	Anomalous Process Creation (Windows Event ID: 4688)
Observable 2	<i>Network Traffic Captures Contain Anomalous Binaries</i>
Observable 3	<i>Network Traffic Captures Contain Anomalous Function Code Related to Shadowpad Plug-Ins</i>
Observable 4	Anomalous Object Access Attempt (Windows Event ID: 4663)
Observable 5	<i>Anomalous Object Deletion (Windows Event ID: 4660)</i>
Observable 6	<i>Anomalous Registry Value Was Modified (Windows Event ID: 4657)</i>
Observable 7	<i>Presence of Anomalous Registry Keys</i>
Observable 8	<i>Presence of Anomalous Registry Keys: HKLM\HKCU\Software\Microsoft\<Key></i>

Observables Associated with Indicator Removal on Host Technique (T0872)	
Observable 1	<i>Presence of Anomalous Encrypted Files</i>
Observable 2	<i>Anomalous File Deletion (Windows Event ID: 4660)</i>
Observable 3	Anomalous Object Access Attempt (Windows Event ID: 4663)
Observable 4	<i>Anomalous Object Deletion (Windows Event ID: 4660)</i>

Observables Associated with Command-Line Interface (T0807)	
Observable 1	<i>Appearance of Anomalous Command-Line Terminal at Startup</i>
Observable 2	<i>Anomalous Process Creation (Windows Event ID: 4688)</i>
Observable 3	<i>Presence of Anomalous Application</i>
Observable 4	Presence of Anomalous Application with Plugin Name
Observable 5	Presence of Anomalous Application with Plugin Name: Shell
Observable 6	Presence of Anomalous Application with Plugin ID
Observable 7	Presence of Anomalous Application with Plugin ID: 0x130
Observable 8	<i>Presence of Application with Anomalous Hash</i>
Observable 9	Presence of Application with Anomalous Hash: 9f9d96e99cef99cbfe8d02899919a7f7220f2273bb36a084642f492dd3e473da

Observables Associated with Loss of Availability Technique (T0826)	
Observable 1	<i>Anomalous Interruption of Processes</i>
Observable 2	<i>Anomalous Interruption of Processes in the Information Technology (IT) Environment</i>
Observable 3	<i>Anomalous Interruption of Business Processes</i>
Observable 4	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment</i>
Observable 5	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment: Supervisory Control and Data Acquisition (SCADA)</i>
Observable 6	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment: Load Dispatch</i>
Observable 7	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment: Power Distribution</i>
Observable 8	<i>Anomalous Non-Responsive Devices</i>
Observable 9	<i>Power Loss For Majority of Customer Base</i>
Observable 10	<i>Power Loss For Majority of Customer Base For 3 Hours</i>
Observable 11	<i>Power Loss For Portion of Customer Base</i>
Observable 12	<i>Power Loss For Portion of Customer Base For 12 Hours</i>
Observable 13	<i>Anomalous Device Alarms</i>
Observable 14	<i>Anomalous Device Alarms in the Information Technology (IT) Environment</i>
Observable 15	<i>Anomalous Device Alarms in the Operational Technology (OT) Environment</i>
Observable 16	<i>Anomalous Device Alarms in the Operational Technology (OT) Environment: Supervisory Control and Data Acquisition (SCADA)</i>
Observable 17	<i>Anomalous Supervisory Control and Data Acquisition (SCADA) Alerts</i>

Observables Associated with Loss of Productivity and Revenue Technique (T0828)	
Observable 1	<i>Anomalous Interruption of Processes</i>
Observable 2	<i>Anomalous Interruption of Processes in the Information Technology (IT) Environment</i>
Observable 3	<i>Anomalous Interruption of Business Processes</i>
Observable 4	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment</i>
Observable 5	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment: Supervisory Control and Data Acquisition (SCADA)</i>
Observable 6	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment: Load Dispatch</i>
Observable 7	<i>Anomalous Interruption of Processes in the Operational Technology (OT) Environment: Power Distribution</i>
Observable 8	<i>Anomalous Non-Responsive Devices</i>
Observable 9	<i>Power Loss For Majority of Customer Base</i>
Observable 10	<i>Power Loss For Majority of Customer Base For 3 Hours</i>
Observable 11	<i>Power Loss For Portion of Customer Base</i>
Observable 12	<i>Power Loss For Portion of Customer Base For 12 Hours</i>
Observable 13	<i>Company Experiences Anomalous Loss of Revenue</i>
Observable 14	<i>Company Experiences Anomalous Loss of Revenue: Maharashtra State Electricity Transmission Company Limited (MSETCL)</i>
Observable 15	<i>Company Experiences Anomalous Loss of Revenue: \$1.8 Million</i>
Observable 16	<i>Company Experiences Anomalous Loss of Revenue: One Day Outage</i>
Observable 17	<i>City Experiences Anomalous Loss of Productivity</i>
Observable 18	<i>City Experiences Anomalous Loss of Productivity: City of Mumbai</i>
Observable 19	<i>City Experiences Anomalous Loss of Productivity: \$1 Billion</i>
Observable 20	<i>City Experiences Anomalous Loss of Productivity: One Day Outage</i>
Observable 21	<i>Capability Loss</i>
Observable 22	<i>Capability Loss: 3508 MW</i>
Observable 23	<i>Generation Loss</i>
Observable 24	<i>Generation Loss: 1375 MW</i>

APPENDIX B: ARTIFACTS LIBRARY

Artifacts Associated with Supply Chain Compromise Technique (T0862)	
Artifact 1	MAC Address
Artifact 2	LLDP Requests
Artifact 3	Usage of Vendor Maintenance Account
Artifact 4	Usage of Default Account
Artifact 5	Static Source IP Address
Artifact 6	Ping Echo Port
Artifact 7	HTTP Port
Artifact 8	SNMP Port
Artifact 9	SMB Port
Artifact 10	Network Discover Protocols
Artifact 11	Domain Name
Artifact 12	Source IP Address
Artifact 13	Mismatched Software Hashes
Artifact 14	DNS Queries Traffic Port
Artifact 15	Inaccurate Delivery Based on Design Documents
Artifact 16	Destination IP Address
Artifact 17	Physical Defects to Hardware
Artifact 18	Factory Acceptance Test Failure
Artifact 19	Inconsistencies in Hardware Bill of Materials
Artifact 20	Inconsistencies in Software Bill of Materials
Artifact 21	Hardware Serial Number Missing
Artifact 22	Unscheduled Firmware Updates
Artifact 23	Domain Registrant Data
Artifact 24	Hardware Failed Site Acceptance Test
Artifact 25	Hardware Tampering Evidence
Artifact 26	Device Incompatibility Issues
Artifact 27	Device Failures
Artifact 28	Additional Hardware Inserted on Devices
Artifact 29	Domain Autonomous System Number
Artifact 30	Domain IP Resolution
Artifact 31	Manipulation of Signature on Digital Certifications

Artifacts Associated with Standard Application Layer Protocol Technique (T0869)	
Artifact 1	SMB Traffic Port
Artifact 2	Network Connection Times
Artifact 3	External IP Addresses
Artifact 4	External Network Connections
Artifact 5	DNS Autonomous System Number
Artifact 6	Increase in the Number of External Connections
Artifact 7	RDP Traffic Port
Artifact 8	HTTP Traffic Port
Artifact 9	DNS Traffic Port
Artifact 10	HTTP Post Request
Artifact 11	HTTPS Traffic Port
Artifact 12	Network Content Metadata

Artifacts Associated with Commonly Used Port Technique (T0885)	
Artifact 1	Unexpected Process Usage of Common Port Observed via Firewall Logs
Artifact 2	Unexpected Process Usage of Common Port Observed via OS Commands (netstat)
Artifact 3	Unexpected Process Usage of Common Port Observed via Memory
Artifact 4	Unexpected Process Usage of Common Port Observed via OS Logs
Artifact 5	Unexpected Host Communicating with Common Port on Industrial Asset

Artifacts Associated with Connection Proxy Technique (T0884)	
Artifact 1	Unexpected Process Usage of Network Proxy Port Observed via Memory
Artifact 2	Unusual Network or Host Communications Identified in Network Proxy Log
Artifact 3	Unexpected Host Communicating with Network Proxy Port on Industrial Asset
Artifact 4	Unexpected Process Usage of Network Proxy Port Observed via OS Logs
Artifact 5	Unexpected Application Communication to Network Proxy Port in Command Line Output (netstat)
Artifact 6	Unexpected Process Usage of Network Proxy Port Observed via Firewall Logs

Artifacts Associated with Program Download (T0843)	
Artifact 1	Controller State Change
Artifact 2	Controller Connection to External Website
Artifact 3	Controller In Stop State
Artifact 4	Controller Connected to External Networks

Artifacts Associated with Program Download (T0843)	
Artifact 5	Network Traffic Creation
Artifact 6	Network Metadata
Artifact 7	External IP Address
Artifact 8	Controller Network Connections via Management Protocol
Artifact 9	Operational Process Shutdown
Artifact 10	External Domain Connection
Artifact 11	Operational Process Restart
Artifact 12	Controller Application Log Type
Artifact 13	Supervisory Workstation Program Download Popup
Artifact 14	Controller Application Log Event
Artifact 15	Device Alarm
Artifact 16	Device Alert
Artifact 17	Operational Database Data Modification
Artifact 18	Controller Application Log Timestamp
Artifact 19	Controller In Program State

Artifacts Associated with Scripting (T0853)	
Artifact 1	Startup Menu Modification
Artifact 2	OS Service Installation
Artifact 3	Registry Modifications
Artifact 4	Network Services Created
Artifact 5	External Network Connections
Artifact 6	Prefetch Files Created
Artifact 7	Executable Files
Artifact 8	System Processes Created
Artifact 9	OS Timeline Event
Artifact 10	System Event Log Creation
Artifact 11	Files Dropped into Directory
Artifact 12	Windows API Event Log

Artifacts Associated with Hooking (T0874)	
Artifact 1	File Modification
Artifact 2	Files Open
Artifact 3	Mismatch Between Memory Resources (.DLL, Files, Sockets) and Disk Resources

Artifacts Associated with Hooking (T0874)	
Artifact 4	Mismatch Parent to Child Processes
Artifact 5	Executable and Linkable Format (ELF) Binaries
Artifact 6	Memory Writes
Artifact 7	Module Load
Artifact 8	Process Performance Mismatched with User Interface At HMI or
Artifact 9	Files Closed
Artifact 10	.DLL Execution
Artifact 11	PE Header

Artifacts Associated with Remote System Information Discovery (T0888)	
Artifact 1	Unexpected Recon Associated Library Calls
Artifact 2	Unexpected Standard Protocol Usage
Artifact 3	Unexpected Recon Associated Command Line Options (Ping Sweep, netstat, etc.)
Artifact 4	Unexpected Recon Associated Child Processes (Ping Sweep, netstat, etc.)
Artifact 5	Exfiltration of Host, Network, and/or System Architecture or Configuration Data
Artifact 6	Compromise and Exfiltration of Data from Asset Information Datastores or Applications
Artifact 7	Unexpected Industrial Protocol Usage
Artifact 8	Unexpected Industrial Application Usage

Artifacts Associated with Masquerading (T0849)	
Artifact 1	Command Line Execution
Artifact 2	Additional Functionality In Applications
Artifact 3	Applications Causing Unintended Actions
Artifact 4	Leetspeak File Creation
Artifact 5	File Modification
Artifact 6	Process Metadata Changes
Artifact 7	Common Application with Non-Native Child Processes
Artifact 8	Scheduled Job Metadata
Artifact 9	Services Metadata
Artifact 10	Service Creation
Artifact 11	Scheduled Job Modification
Artifact 12	Additional File Directories Created
Artifact 13	File Creation with Common Name

Artifacts Associated with Masquerading (T0849)

Artifact 14	Leetspeak User Metadata
Artifact 15	Warez Application Use

Artifacts Associated with Execution through API (T0871)

Artifact 1	Vendor Specific Network Traffic
Artifact 2	Function Execution
Artifact 3	SCADA Protocol Network Traffic
Artifact 4	Data Sent with Large File Size
Artifact 5	Data Received with Large File Size
Artifact 6	Network Traffic with Command Execution Content
Artifact 7	State Change in The Process
Artifact 8	Industrial Network Traffic
Artifact 9	Remote Connections
Artifact 10	IP Addresses from Network Traffic
Artifact 11	Controller Failure
Artifact 12	Timestamps Associated with Activity
Artifact 13	Controller Configuration Change
Artifact 14	Common Network Traffic
Artifact 15	API Log Event (If Enabled)
Artifact 16	Module Load
Artifact 17	Reboot
Artifact 18	Process Failure
Artifact 19	Control Logic Change

Artifacts Associated with Indicator Removal on Host (T0872)

Artifact 1	HMI Dialog Box Open
Artifact 2	API System Calls
Artifact 3	HMI Interface Manipulation
Artifact 4	Process Creation
Artifact 5	Command Execution
Artifact 6	File Creation
Artifact 7	HMI Dialog Box Close
Artifact 8	User Logon Event
Artifact 9	Windows Registry Key Modification

Artifacts Associated with Indicator Removal on Host (T0872)	
Artifact 10	Windows Registry Key Deletion
Artifact 11	User Logoff Event
Artifact 12	HMI Screen Changes
Artifact 13	Missing Log Events
Artifact 14	Unexpected Reboots
Artifact 15	Windows Security Log 1102 for Cleared Events
Artifact 16	File Deletion
Artifact 17	File Modification
Artifact 18	Sdelete Executable Loaded
Artifact 19	Sdelete Executable Executed
Artifact 20	File Metadata Changes
Artifact 21	Timestamp Inconsistencies
Artifact 22	User Authentication
Artifact 23	Memory Writes

Artifacts Associated with Command-Line Interface (T0807)	
Artifact 1	Command Execution
Artifact 2	Application Log
Artifact 3	HTTP Traffic
Artifact 4	Telnet Traffic
Artifact 5	SSH Traffic
Artifact 6	VNC Traffic Port
Artifact 7	Process Creation
Artifact 8	Remote Connections
Artifact 9	Process Ending
Artifact 10	Script Execution
Artifact 11	User Account Logon
Artifact 12	User Account Privilege Change
Artifact 13	Logon Event
Artifact 14	Event Log Type
Artifact 15	Event Log Type
Artifact 16	Failed Logon Event
Artifact 17	Command Line Memory Data
Artifact 18	cmd.exe Application Execution

Artifact 19	RDP Traffic
Artifact 20	Industrial Application Execution
Artifact 21	POWERSHELL Cmdlet Application Execution
Artifact 22	Event ID 4103 POWERSHELL Command
Artifact 23	Event ID 4688 Command Line Execution
Artifact 24	NTUSER Application Execution Entries
Artifact 25	External Network Connection

Artifacts Associated with Loss of Availability (T0826)	
Artifact 1	Process Failure Due to Loss of Required Network or System Dependency
Artifact 2	Unexplained Loss of User Data
Artifact 3	Changes in Network Routing or Usage of Redundant Control System Network Connection Due to Failed Network Path
Artifact 4	Significant Reduction or Increase in Network Traffic Due to Malware Propagation or Disappearance of Services
Artifact 5	Significant Logged Usage of Native Crypto Functions or Presence of Import of Crypto Functions in Binaries
Artifact 6	Operator or User Discovery of Encrypted or Inoperable Systems
Artifact 7	File System Modification Artifacts Might Be Associated with the Loss of Availability Might Be Present on Disk
Artifact 8	Unexplained Loss of Application Data

Artifacts Associated with Loss of Productivity and Revenue (T0828)	
Artifact 1	Loss of Confidence in a Safety System Due to Unreliability Might Result in a Risk Management Driven Shutdown of a Plant
Artifact 2	Wormable or Other Highly Propagating Malware Might Result in the Shutdown of a Plant to Prevent Ransomware or Other Destructive Attacks
Artifact 3	Extortion Attempts Might Lead to Reduced Operations Due to Potential Presence of Malicious Attackers
Artifact 4	Loss of Control of Critical Systems Due to Ransomware or Loss of Confidence Might Lead to a Degraded Productivity or Revenue Operating State
Artifact 5	File System Modification Artifacts Might Be Associated with the Loss of Productivity and Revenue Attack Might Be Present on Disk

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist • Security Tester
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

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